SE Lab 1
PROBLEM STATEMENT #23 | Smart Cities, Transport & Logistics
Hyperlocal Courier Dispatch & Tracking Engine
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Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall match an outgoing courier request with the nearest active delivery rider within a 3 km radius.	High	Pass: Rider receives dispatch notification with pickup location. Fail: Order assigned to an offline rider.	Given by problem statement — core dispatch function that connects senders to available riders quickly.	
FR-002	Functional	The system shall allow the Sender Client to view the assigned rider's current location and shipment status on a map during an active delivery.	High	Pass: the sender can view the assigned rider's current location and shipment status on the map during an active delivery. Fail: the rider's location or shipment status cannot be displayed during an active delivery.	Real-time visibility builds sender trust and reduces support inquiries about shipment location.	
FR-003	Functional	The system shall allow the Delivery Rider to accept or reject a dispatched pickup request within 60 seconds of notification.	High	Pass: rider taps Accept/Reject and, on Reject or timeout, the system reassigns the request to the next-nearest rider. Fail: request remains pending with no reassignment.	A hard response window prevents delivery delays caused by unresponsive riders.	
FR-004	Functional	The system shall compute and display an optimized multi-stop route for a rider carrying more than one active parcel.	Medium	Pass: generated stop sequence has total travel distance less than or equal to the unoptimized (address-order) route for the same stop set. Fail: route omits one or more active stops.	Route optimization reduces rider travel time and fuel cost when handling multiple parcels at once.	
FR-005	Functional	The system shall require the Delivery Rider to enter a One-Time Password (OTP), provided to the recipient, before a delivery can be marked complete.	High	Pass: delivery status changes to "Delivered" only after the correct OTP is entered. Fail: delivery can be marked complete without OTP entry.	OTP verification confirms physical handover to the correct recipient and prevents fraudulent completions.	
NFR-001	Performance & Security	Delivery status and live GPS telemetry updates from riders shall transmit to the Sender Client with under 2-second latency.	High	Pass: benchmarking tests confirm target latency and security standards are met under simulated peak load.	Given by problem statement — low-latency, secure location updates are central to a live tracking experience.	
NFR-002	Reliability & Availability	The dispatch-matching service shall maintain at least 99.5% uptime, measured on a rolling 30-day basis.	High	Pass: uptime monitoring logs show availability of 99.5% or higher over a rolling 30-day window. Fail: downtime exceeds the 0.5% threshold.	Courier dispatch is time-critical; any downtime directly blocks new pickups and stalls active deliveries.	